Jiajia Yu

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Research interests

Overall — applied and computational math, mathematics of data science/machine learning.

Current — mean-field games/control and optimal transport.

Employment

Aug. 2023 – July 2026 — **Duke University** – Durham, NC
Phillip Griffiths Assistant Research Professor
Mentors: Prof. Xiuyuan Cheng, Prof. Jian-Guo Liu and Prof. Hongkai Zhao

Sept. 2018 – May 2023 — **Rensselaer Polytechnic Institute** – Troy, NY
Research Assistant and Teaching Assistant

Education

2018 – 2023 — **Rensselaer Polytechnic Institute** – Troy, NY
Ph.D. in Mathematics. *GPA: 4/4.* Mentor: Prof. Rongjie Lai

2013 – 2017 — **Beijing Normal University** – Beijing, China
B.S. in Mathematics and Applied Mathematics. *Major GPA: 96/100*

Publications

Preprints

[4] Han Huang, **Jiajia Yu**, Tianyi Chen, Rongjie Lai, *Joint Inference of Trajectory and Obstacle in Mean-Field Games via Bilevel Optimization*, arXiv:2507.19344, 2025. (Submitted)

[3] **Jiajia Yu**, Xiuyuan Cheng, Jian-Guo Liu, Hongkai Zhao, *Convergence Analysis and Acceleration of Fictitious Play for General Mean-Field Games via Best Response*, arXiv:2411.07989, 2024. (Submitted)

[2] Yu Liu, Weibin Peng, Tianyu Wang, **Jiajia Yu**, *Zeroth-order Stochastic Cubic Newton Method Revisited*, arXiv:2410.22357, 2024. (Submitted)

[1] Tianyu Wang, Zicheng Wang, **Jiajia Yu**, *Zeroth-order Low-rank Hessian Estimation via Matrix Recovery*, arXiv:2402.05385, 2024. (Submitted)

Journal

[5] **Jiajia Yu**, Jian-Guo Liu, Hongkai Zhao, *Equilibrium Correction Iteration for A Class of Mean-Field Game Inverse Problem*, Inverse Problems, **41** (2025) 125009 https://doi.org/10.1088/1361-6420/ae2b09

[4] **Jiajia Yu**, Quan Xiao, Tianyi Chen, Rongjie Lai, *A Bilevel Optimization Approach for Inverse Mean-Field Games*, Inverse Problems, **40** (2024) 105016 https://doi.org/10.1088/1361-6420/ad75b0

[3] **Jiajia Yu**, Rongjie Lai, Wuchen Li, Stanley Osher, *A Fast Proximal Gradient Method and Convergence Analysis for Dynamic Mean Field Planning*, Mathematics of Computation, 93 (2024), 603-642. https://doi.org/10.1090/mcom/3879

[2] Han Huang, **Jiajia Yu**, Jie Chen, Rongjie Lai, *Bridging Mean-Field Games and Normalizing Flows with Trajectory Regularization*, Journal of Computational Physics, Vol. 487, 112155, 2023. https://doi.org/10.1016/j.jcp.2023.112155

[1] **Jiajia Yu**, Rongjie Lai, Wuchen Li, Stanley Osher, *Computational Mean-field Games on Manifolds*, Journal of Computational Physics, Vol. 484, 112070, 2023. https://doi.org/10.1016/j.jcp.2023.112070

Conference

[1] **Jiajia Yu**, Junghwan Lee, Yao Xie, Xiuyuan Cheng, *High-dimensional Mean-Field Games by Particle-based Flow Matching*, arXiv:2512.01172, 2025. (Extended version accepted by ICLR 2026. Short version accepted by NeurIPS 2025 Workshop DynaFront.)

Awards

2024 **SIAM Early Career Travel Award**, MDS24, SIAM

2023 **Karen and Lester Gerhardt Prize**, School of Science, RPI

2023 **Joaquin B. Diaz Memorial Prize**, Department of Mathematical Sciences, RPI

2022 **AWM Travel Grant**, AWM

Teaching

Instructor, Duke University, Durham, NC

2025F Math221&721 Linear Algebra

2024S, 2025S Math466&766 Mathematics of Machine Learning

2023F, 2024F Math465&765/COMPSCI445/STA465 Introduction to High-Dimensional Data Analysis (2023F co-teach with Prof. Xiuyuan Cheng)

Teaching Assistant, Rensselaer Polytechnic Institute, Troy, NY

2019F MATH 4400 ODE and Dynamical Systems, Instructor: Prof. Gregor Kovačič

2019F MATH 4200 Mathematical Analysis I, Instructor: Prof. Bruce Piper

2019S MATH 4020 Introduction to Number Theory, Instructor: Prof. Bruce Piper

2018F MATH 4200 Mathematical Analysis I, Instructor: Prof. Fengyan Li

2018F MATH 4040 Introduction to Topology, Instructor: Prof. Bruce Piper

Mentoring

May 2025 – July 2025 **Math+ Program (Undergraduate Research Project)**
Project: Translation-invariant optimal transport distance. (Manager: Shrikant Chand)
Students: Peilin He, Zakk Heile, Jayson Tran, Alice Wang. (All are rising juniors at Duke.)
Outcome: *Rigid Invariant Sliced Wasserstein via Independent Embeddings*, arXiv:2510.10233. Short version accepted by NeurIPS2025 workshop. Extended version submitted to ICLR2026.

Presentations

Invited Seminar Talks

Feb. 2026 Learning in mean-field games, *Level Set Meeting,* UCLA (Virtual).

Jan. 2026 Learning in mean-field games, *Department Seminar,* University of Delaware.

Jan. 2026 Learning in mean-field games, *Department Seminar,* Pennsylvania State University.

Nov. 2025 Learning and inference in mean-field games, *Applied and Analysis Seminar,* Duke University.

Oct. 2025 Learning and inference in mean-field games
PSU-Purdue-UMD Joint Seminar on Mathematical Data Science (Virtual).

Oct. 2023 Computational mean-field games: from conventional methods to deep generative models
IMA Data Science Seminar, University of Minnesota.

Oct. 2023 A bilevel optimization approach for inverse mean-field games
RTG Seminar, University of South Carolina.

July 2023 Computational mean-field games: from conventional methods to deep generative models
Summer School on Mathematical Foundation of Data Science, University of South Carolina (Virtual).

June 2022 Computational mean-field games on manifolds
Optimal Transport and Mean-Field Games Seminar, University of South Carolina (Virtual).

March 2021 An efficient and flexible algorithm for dynamic mean-field planning and convergence analysis
Optimal Transport and Mean-Field Games Seminar, University of South Carolina (Virtual).

Invited Workshop/Workshop Talks

Jan. 2026 Learning and inference in mean-field games, Westlake University (Virtual).

July 2025 Learning and inference in mean-field games
Sampling, Inference, and Data-Driven Physical Modeling in Scientific Machine Learning, IPAM, CA.

Aug. 2024 Computational methods for the mean-field game and its inverse game
Theory and Applications for Optimal Control and Generative Models, Purdue University.

July 2024 *Empowering a Diverse Computational Mathematics Research Community,* ICERM, RI.

May. 2023 *AMS MRC Conference: Ricci Curvatures of Graphs and Applications to Data Science,* Beaver Hollow Conference Center, Java Center, NY.

Conference Talks

Mar. 2025 Convergence Analysis and Acceleration of Fictitious Play for General Mean-Field Games via the Best Response, *AMS 2025 Spring Central Sectional Meeting,* University of Kansas.

Jan. 2025 Bridging mean-field games and normalizing flows with trajectory regularization, *JMM 2025,* Seattle, WA.

Oct. 2024 Computational methods for inverse mean-field games, *SIAM MDS24,* Atlanta, GA.

May. 2024 A bilevel optimization approach for inverse mean-field games, *SIAM IS24,* Atlanta, GA.

Oct. 2023 Computational mean-field games on manifolds, *SIAM NYNJPA 1st Annual Meeting,* NJIT.

Aug. 2023 A bilevel optimization approach for inverse mean-field games, *MOPTA,* Lehigh University.

Posters

Feb. 2026 — High-dimensional Mean-Field Games by Particle-based Flow Matching, *Algorithmic Advances in Generative AI,* Duke University.

Dec. 2025 — High-dimensional Mean-Field Games by Particle-based Flow Matching, *Dynamics at the Frontiers of Optimization, Sampling, and Games,* Workshop at NeurIPS 2025, San Diego.

May 2025 — Convergence Analysis and Acceleration of Fictitious Play for General Mean-Field Games via the Best Response, *NSF CompMath Meeting 2025,* University of Utah.

Mar. 2025 — Convergence Analysis and Acceleration of Fictitious Play for General Mean-Field Games via the Best Response, *Statistics and Optimal Transport Workshop,* Columbia University.

Nov. 2023 — A bilevel optimization approach for inverse mean-field games
Triangle Computational and Applied Mathematics Symposium, Duke University.

June 2022 — Computational mean-field games on Euclidean space and manifolds
The 2022 AWM Research Symposium Poster Session, University of Minnesota.

Professional Services

Conference Organization

Oct. 2024 — Mini-symposium at SIAM MDS24, Atlanta, GA.
Incorporating Optimal Transport in Machine Learning, co-organize with Alex Cloninger (UCSD).

Oct. 2023 — Mini-symposium at SIAM NYNJPA 1st Annual Meeting, NJIT, Newark, NJ.
Optimal Transport: Computation, Applications, and Extensions, co-organize with Rongjie Lai (Purdue).

Journal/Book/Conference Reviewer

Journal of Computational Physics (JCP), Multiscale Modeling and Simulation (MMS), SIAM Imaging Science (SIIMS), SIAM Applied Mathematics (SIAP), Journal of Scientific Computing (JSC), Inverse Problems, Research in the Mathematical Sciences.

Advances in Data Science.

Neurips Workshop.

Outreach

2024 — Judge for Triangle Competition in Mathematical Modeling (TriCoMM)

2022 – 2023 — Vice President of AWM Student Chapter at RPI

Recommenders

Ph.D. Advisor — Rongjie Lai, lairj@purdue.edu
Professor, Purdue University, Department of Mathematics.

Teaching — Clark Bray, cbray@duke.edu
Associate Professor of the Practice of Mathematics, Duke University, Department of Mathematics.

Research Xiuyuan Cheng, xiuyuan.cheng@duke.edu
Professor, Duke University, Department of Mathematics.

Research Alexander Cloninger, acloninger@ucsd.edu
Professor, University of California, San Diego, Department of Mathematics and Halicioğlu Data Science Institute (HDSI).

Research Jian-Guo Liu, jian-guo.liu@duke.edu
Professor, Duke University, Department of Mathematics and Department of Physics.

Research Hongkai Zhao, hongkai.zhao@duke.edu
Professor, Duke University, Department of Mathematics.